

Sungmoon Choi

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RESEARCH INTERESTS

Astrodynamics, Orbital Mechanics, Space Mission Design and Navigation, Trajectory Optimization, Tour Design, Multi-Body Dynamics, Global Optimization, Machine Learning, Artificial Intelligence, Contour Integral

EDUCATION

Ph.D. , <i>Aerospace Engineering (Minor: Computer Science)</i>	Iowa State University, Ames, IA	2026.05
M.S. , <i>Aerospace Engineering</i>	Iowa State University, Ames, IA	2025.12
M.S. , <i>Astronomy</i>	Yonsei University, Seoul, Korea	2020.02
B.S. , <i>Mechanical Engineering</i>	Pusan National University, Busan, Korea	2016.08

PROFESSIONAL APPOINTMENTS

New Mexico State University	Las Cruces, NM
Assistant Professor (<i>Tenure-Track</i>), Department of Mechanical and Aerospace Engineering	2026.08 – Present
NASA Jet Propulsion Laboratory	Pasadena, CA
Research Intern , Mission Design and Navigation Section	2023.06 – 2025.12

AWARDS & ACHIEVEMENTS

• Miller Completion Award, <i>Iowa State University</i>	2025.12
• ISEB-KARI Scholarship awarded by the Space Generation Advisory Council (<i>SGAC</i>)	2025.09
• Research Excellence Award, <i>Iowa State University</i>	2025.05
• Exceptional Performance Award for GTOC Team, <i>NASA JPL</i>	2023.09
• 1st Place at Global Trajectory Optimization Competition (GTOC) 12, <i>NASA JPL</i>	2023.08
• Top 5 Proposal as Principal Investigator, <i>NASA L'SPACE Program</i>	2022.08
• Outstanding Service Award, <i>Korean-American Scientists and Engineers Association (KSEA)</i>	2022.08
• Excellent Paper Presentation Award, <i>Korean Society for Aeronautical and Space Sciences</i>	2019.11
• IAC / ISEB / SGAC 2017 Representative of KARI, <i>Korea Aerospace Research Institute (KARI)</i>	2017.09

SCHOLARSHIPS & FELLOWSHIPS

- The Korean Government Scholarship Program for Study Overseas
- Research / Teaching Assistantship from graduate program
- OK Savings Bank Fellowship
- Academic Excellence Scholarship
- National Scholarship
- Education capability enhancement project

JOURNAL PAPERS

- 2026 (J3) **Choi, S.**, Das-Stuart, A., Anderson, R. L., & Abdelkhalik, O. (2026). Vectorized dynamic-size genetic algorithm for rapid generation of optimal and suboptimal flyby sequences. *Astrodynamics*, 1-24. <https://doi.org/10.1007/s42064-026-0317-5>
- (J2) **Choi, S.**, Negrete, A., Abdelkhalik, O., Servadio, S., & Lee, D. D. (2026). Ambiguity-Free Geometric Approach to Deep-Space Navigation Using Angles-Only Measurements. *Journal of Guidance, Control, and Dynamics*, 1-14. <https://doi.org/10.2514/1.G009699>
- 2025 (J1) **Choi, S.**, Abdelkhalik, O., & He, P. (2025). Multidisciplinary optimization of end-to-end Mars aerobraking trajectory and spacecraft design. *Advances in Space Research* 75, no. 11 (2025): 8065-8083. <http://dx.doi.org/10.1016/j.asr.2025.03.022>

CONFERENCE PROCEEDINGS

- 2026 (C12) **Choi, S.**, Abdelkhalik, O., Servadio, S., & Lee, D. (2026). Deep-Space Initial Orbit Determination from Angles-Only Measurements Without Prior Attitude or Distance Information. In *AIAA/AAS Astrodynamics Specialist Conference*
- (C11) **Choi, S.**, Gibson, T., & Abdelkhalik, O. (2025). Learning-Based Placement of Deep Space Maneuvers in the Rendezvous Problem. In *Daniele Mortari Astronautics Symposium (URL)*
- (C10) Negrete, A., **Choi, S.**, & Abdelkhalik, O. (2025). Shape-based Approximation for Low-Thrust Spacecraft Trajectory Using Chebyshev Polynomials. In *Daniele Mortari Astronautics Symposium (URL)*
- 2025 (C9) **Choi, S.**, & Abdelkhalik, O. (2025). Shape-based Low-Thrust ΔV Prediction Using Deep Neural Networks. In *Daniele Mortari Astronautics Symposium (URL)*
- (C8) **Choi, S.**, Das-Stuart, A., Anderson, R. L., & Abdelkhalik, O. (2025). Reinforcement Learning for Multiple Gravity Assist Trajectory Optimization. In *76th International Astronautical Congress (IAC 2025)*
- (C7) **Choi, S.**, Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2025). Neural Network-Based Universal Variable Approximation for Lambert's Problem. In *AIAA/AAS Space Flight Mechanics Meeting*
- (C6) **Choi, S.**, Abdelkhalik, O., & He, P. (2025). Multidisciplinary Optimization of End-to-End Mars Aerobraking Trajectory and Spacecraft Design. In *AIAA/AAS Space Flight Mechanics Meeting*
- (C5) **Choi, S.**, Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2024). Vectorized Dynamic Size Genetic Algorithm for Fast Flyby Sequence Generation. In *AIAA/AAS Astrodynamics Specialist Conference*
- 2024 (C4) **Choi, S.**, Abdelkhalik, O., Das-Stuart, A., & Anderson, R. L. (2024). Machine Learning Application for Lambert's Problem. In *AIAA/AAS Astrodynamics Specialist Conference*
- (C3) **Choi, S.**, Knapick, D., Vijayakumar, M., & Abdelkhalik, O. (2024). Hidden Genes Genetic Algorithm with Low-Thrust Shape-Based Method. In *AIAA SCITECH 2024 Forum* (p. 0629)
- 2019 (C2) **Choi, S.**, Choi, J., & Park, C. (2019). Trans-Lunar Trajectory Design Using Virtual Orbital Plane Based on Ephemeris. *Abstracts of the Korean Society for Aeronautical and Space Sciences Conference*, 214-215
Excellent Paper Presentation Award
- 2018 (C1) Choi, J., Hwang, J., **Choi, S.**, & Park, C. (2018). Preliminary Mission Planning for Multiple Asteroid Exploration. *Abstracts of the Korean Society for Aeronautical and Space Sciences Conference*, 313-314

SELECTED PRESENTATIONS

- 26.04.10 **Choi, S.** "Machine Learning-Aided Tour Design and Angles-Only Navigation for Deep-Space Missions," Seminar in the Department of Mechanical and Aerospace Engineering, New Mexico State University, *Las Cruces, NM*
- 26.03.05 **Choi, S.** "Machine Learning-Aided Tour Design and Trajectory Optimization," Seminar in the Department of Mechanical and Aerospace Engineering, California State University, Long Beach, *Long Beach, CA*
- 25.12.18 **Choi, S.** "Reinforcement Learning for Multiple Gravity Assist Trajectory Optimization," Final Presentation in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*
- 25.08.06 **Choi, S.** "Machine Learning 101," Invited Lecture in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*
- 24.08.05 **Choi, S.** "Neural Network-Based Lambert's Problem & Vectorized Dynamic-Size Genetic Algorithm," Final Presentation in Mission Design and Navigation Section at Jet Propulsion Laboratory, *Pasadena, CA*

POSTER PRESENTATIONS

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| 2026.07 | (P4) Yi, A., Servadio, S., & Choi, S. (2026). Robust Deep-Space Navigation Via Four-Body Angles-Only Analysis. EURECA Summer Undergraduate Research Showcase, Ames, Iowa |
| 2025.04 | (P3) Choi, S. , & Abdelkhalik, O. (2025). Ambiguity-Free Geometrical Deep Space Navigation Using Angle-Only Measurements. Iowa Space Grant Consortium Spring Research Symposium, Ames, Iowa |
| 2019.02 | (P2) Choi, S. , & Park, C. (2019). That's One Small Step for a Man, One Giant Leap for Mankind. The 2 nd Astro Poster Jamboree. Seoul, Korea |
| 2014.08 | (P1) Choi, S. , Park, H., Park, J., Chung, Y., Held, D., & Kim, D. (2014). Development of Smart Traps for Asian Ambrosia Beetles. UKC 2014, San Francisco, CA |

STUDENT CONFERENCE PROCEEDINGS

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| 2026 | (S3) Larsen, K. (2026, March). A Contour-Integral Approach to Collinear Lagrange Points and Zero-Velocity Region Openings in the Circular Restricted Three-Body Problem. In <i>2026 Regional Student Conferences</i> (p. 114205).
(S2) Roy, T. (2026, March). Low-Thrust Trajectory Optimization via Direct Transcription Using Extreme Theory of Functional Connections. In <i>2026 Regional Student Conferences</i> (p. 114788).
(S1) Park, M. (2026, March). A Comparative Study of Physics-Informed Neural Networks and Extreme Theory of Functional Connections for Two-Point Orbital Transfers. In <i>2026 Regional Student Conferences</i> (p. 115555). |
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CURRENT RESEARCH PROJECTS

Reinforcement Learning for Tour Design

- Developing tour design algorithms and contingency planning with trajectory reuse, in collaboration with NASA JPL

Machine Learning Applications to Lambert's Problem

- Constructing data-driven frameworks for fast prediction of Lambert's problems, in collaboration with NASA JPL

Geometric Approach to Optical Navigation for Deep Space Missions

- Integrating unified position/attitude measurement models into state estimation, in collaboration with Iowa State University

Low-Thrust Trajectory Design Using a Shape-Based Approach

- Establishing shape-based trajectory formulation incorporating necessary conditions of optimality

Machine Learning-Based Prediction of Optimal Deep Space Maneuvers

- Applying machine learning models to predict the location and timing of optimal deep space maneuvers

Physics-Informed Neural Network Application to the Circular Restricted Three-Body Problem

- Investigating physics-based machine learning to resolve two-point boundary-value problems in CRTBP

Low-Thrust Trajectory Optimization Using Extreme Theory of Functional Connections

- Leveraging extreme theory of functional connection for efficient solution of low-thrust optimal control problems

Contour-Integral Methods in the Circular Restricted Three-Body Problem

- Computing families of periodic orbits in the CRTBP via contour-integral techniques

ADVISING AND MENTORING

Undergraduate Students

• Tishyo Roy , <i>Iowa State University</i> – (S2)	2025.09 – Present
• Mingeun Park , <i>Iowa State University</i> – (S1)	2024.12 – Present
• Kevin Vo , <i>University of Minnesota</i>	2026.04 – Present
• Andrew Yi , <i>University at Buffalo</i> – (P4)	2026.06 – 2026.08
• Kimberly Larsen , <i>Iowa State University</i> – (S3)	2025.09 – 2026.05
• Ethan Griesman , <i>Iowa State University</i>	2026.04 – 2026.06
• Samantha Austin (<u>Boeing Research Fellow</u>), <i>Iowa State University</i>	2025.10 – 2025.12
• Timothy Gibson , <i>Iowa State University</i> – (C11)	2025.02 – 2025.12
• Nicholas Fang , <i>Iowa State University</i>	2025.02 – 2025.05
• Akshatra Sureshababu , <i>Iowa State University</i>	2024.09 – 2025.09
• Modi Mihir (<u>Boeing Research Fellow</u>), <i>Iowa State University</i>	2023.09 – 2024.06

TEACHING EXPERIENCE

Instructor, New Mexico State University

- Upcoming courses: Orbital Mechanics (*AE 362*) / Aerospace System Engineering (*AE 424*)

Teaching Assistant, Iowa State University

- Mechanics of Materials Laboratory at Iowa State University (*EM 3270*) S2026, F2025
- Flight Control System at Iowa State University (*AERE 3310*) S2025, F2024, S2024, S2023

Teaching Assistant, Yonsei University

- Understanding of Space at Yonsei University (*UCC 1160*) F2019, S2019, F2018, S2018

SERVICE TO PROFESSION

Reviewer for IEEE Transactions on Aerospace and Electronic Systems
 Reviewer for Astrodynamics Journal
 Reviewer for IEEE Congress on Evolutionary Computation
 Reviewer for AIAA SciTech